

A continuous valuation with rational value group

Disproof of Conjecture 00000005754

Abstract

The Hahn-series field with rational coefficients and rational exponents carries a continuous valuation whose intrinsic value group is the additive group $\mathbb{Q}$. This group is not isomorphic to $\mathbb{Z}^I$ for any index set I , even without requiring preservation of order. Consequently no choice of rank can make the conjecture's asserted integer-power description hold. The Lean formalization constructs the valuation, proves continuity for two explicit codomain topologies, identifies the group of values actually attained, and proves the obstruction for every index type.

1 Claim being disproved and conventions

The final clause of the supplied English statement says that “the value group of a continuous valuation is a power of $\mathbb{Z}$, with exponent the rank.” The Chinese statement makes the same assertion. This is a universal assertion about continuous valuations, and it is a conjunct of the conjecture. One continuous valuation whose value group is no integer power therefore suffices to disprove the conjecture as written.

The statement gives neither a topology nor a discreteness hypothesis. We use the standard valuation topology on the domain field. We prove continuity both for the usual order topology on the extended value group and for the finer topology that isolates finite additive values and has tails as neighborhoods of infinity. Thus the counterexample does not rely on assigning the domain an arbitrary discrete topology. A field is, in particular, a ring of the kind mentioned in the statement.

We use additive valuation notation. An additive valuation on a field K is a function

$$w : K \longrightarrow \Gamma \cup \{\infty\}$$

into a linearly ordered abelian group with a largest element adjoined, satisfying

$$w(0) = \infty, \quad w(1) = 0, \quad w(fg) = w(f) + w(g), \quad w(f + g) \geq \min\{w(f), w(g)\}.$$

Here $\infty + \gamma = \infty$. Its intrinsic additive value group is the image of $K^\times$ under w , with the induced addition and order. In particular, an unnecessarily large target group does not enlarge this intrinsic group.

No interpretation of the unspecified conversion formula or of the asserted rank jumps is needed. We will prove the stronger statement that the intrinsic group is not $\mathbb{Z}^I$ for *any* index set I . In particular, it cannot be $\mathbb{Z}^r$ when r is whatever rank is intended in the statement. When $r \in \mathbb{N}$, the corresponding Lean type is `Fin r → ℤ`.

2 The field and its valuation

Let K be the Hahn-series field

$$K = \mathbb{Q}((t^{\mathbb{Q}})).$$

Its elements are formal sums

$$f = \sum_{q \in \mathbb{Q}} a_q t^q, \quad a_q \in \mathbb{Q},$$

whose support $\{q : a_q \neq 0\}$ is well ordered. Addition is coefficientwise and multiplication is convolution. The standard Hahn-series construction makes this a field. In particular, monomials t^q are nonzero, satisfy $t^q t^r = t^{q+r}$, and have inverse t^{-q} . The field structure, including the general inversion construction, is provided and proved by Mathlib's `HahnSeries.instField` in `Mathlib/RingTheory/HahnSeries/Summable.lean`.

Define

$$w(0) = \infty, \quad w(f) = \min \text{supp}(f) \quad (f \neq 0).$$

The minimum exists by well ordering. In a product of two nonzero series, the coefficient at $w(f) + w(g)$ is the product of the two leading coefficients. It is nonzero, and no smaller exponent occurs. Therefore $w(fg) = w(f) + w(g)$. Addition cannot create an exponent below both leading exponents, so $w(f + g) \geq \min\{w(f), w(g)\}$. Also $w(1) = 0$. This is the additive valuation `HahnSeries.addVal` $\mathbb{Q} \rightarrow \mathbb{Q} \cup \{\infty\}$ in the formalization.

For every $q \in \mathbb{Q}$,

$$t^q \neq 0, \quad w(t^q) = q. \tag{1}$$

Thus every rational value is actually attained on a nonzero field element.

3 Topology and continuity

The valuation topology on K has the following neighborhood basis at $a \in K$:

$$B_\gamma(a) = \{f \in K : w(f - a) > \gamma\}, \quad \gamma \in \mathbb{Q}.$$

This is the canonical Hausdorff topological-field structure associated with the valuation. The Lean instance is built with `Valued.mk'`; its topological division-ring and Hausdorff instances are also checked in `Inspect.lean`.

First give $\mathbb{Q} \cup \{\infty\}$ the topology that isolates every rational value and has $\{q : q > \gamma\} \cup \{\infty\}$ as a neighborhood basis at ∞ .

Lemma 1. *The valuation w is continuous for this topology on its codomain.*

Proof. At zero, the inverse image of the indicated tail is exactly $B_\gamma(0)$. For $f \neq 0$, put $r = w(f)$. If $g \in B_r(f)$, then $w(g - f) > r$. The valuation inequality gives $w(g) \geq r$. If $w(g) > r$, applying the same inequality to $f = g - (g - f)$ would give $w(f) > r$, a contradiction. Therefore $w(g) = r$ throughout $B_r(f)$. The valuation is locally constant at every nonzero point, and hence continuous there. $\square$

The ordinary order topology on $\mathbb{Q} \cup \{\infty\}$ is coarser than this topology. Indeed, finite points are isolated in the finer topology, and order neighborhoods of infinity contain tails. Hence w is continuous also for the ordinary order topology.

Mathlib's multiplicative notation reverses the additive order: in Lean the extended target is

$$\text{ExtendedValue} = \text{Multiplicative} (\text{WithTop } \mathbb{Q})^{\text{od}}.$$

Its multiplicative zero denotes additive infinity, and multiplication denotes addition of exponents. The definition `valuation` is exactly `additiveValuation.toValuation`; it does not alter the underlying values. The finer codomain topology is `WithZeroTopology`. The formal theorem `valuation_continuous` uses this topology, while `valuation_continuous_orderTopology` explicitly uses `Preorder.topology ExtendedValue`. Reversing a linear order interchanges the two kinds of open rays, and therefore leaves the ordinary order topology unchanged. These theorems thus establish the two continuity statements above in the multiplicative encoding.

4 The intrinsic value group

The formalization defines an additive-group homomorphism

$$\text{valueHom} : \text{Additive}(K^\times) \longrightarrow \mathbb{Q}, \quad u \longmapsto \min \text{supp}(u).$$

The additive wrapper means that multiplication of units is regarded as addition. The identity $w(uv) = w(u) + w(v)$ is exactly the homomorphism law. The intrinsic group is defined as

$$G = \text{range}(\text{valueHom}) \leq (\mathbb{Q}, +),$$

called `valueGroup` in Lean.

The theorem `valueHom_agrees` identifies this homomorphism with the finite part of the valuation on units. The theorem `mem_valueGroup_iff` gives the following connection to nonzero field elements:

$$q \in G \iff \exists f \in K, f \neq 0 \text{ and } w(f) = q.$$

Equation (1) proves $G = \mathbb{Q}$, formally `valueGroup_eq_top`. The resulting additive isomorphism is `valueGroupEquiv`. Thus the subsequent group obstruction concerns the actual image group of the constructed valuation.

5 No integer power is possible

For an index set I , write $\mathbb{Z}^I$ for the group of functions from I to $\mathbb{Z}$ with pointwise addition.

Proposition 1. *For every index set I , there is no additive-group isomorphism $\mathbb{Q} \cong \mathbb{Z}^I$.*

Proof. Suppose $e : \mathbb{Q} \rightarrow \mathbb{Z}^I$ were such an isomorphism. If I is empty, $\mathbb{Z}^I$ is the trivial group, whereas $0 \neq 1$ in $\mathbb{Q}$, contradicting injectivity.

If I is nonempty, fix $i \in I$ and let $\mathbf{1}$ denote the function that is identically one. Set $x = e^{-1}(\mathbf{1})$ and $y = x/2$. Since $y + y = x$, additivity gives

$$e(y) + e(y) = \mathbf{1}.$$

Evaluating at i yields $2e(y)(i) = 1$ in $\mathbb{Z}$, which is impossible. This contradiction proves the assertion. $\square$

The formal theorem `rationals_not_integer_power` proves Proposition 1 for an arbitrary Lean index type. Composing a proposed isomorphism $G \cong \mathbb{Z}^I$ with `valueGroupEquiv` then proves `valueGroup_not_integer_power`. This rules out even abstract additive isomorphisms, so it also rules out ordered-group isomorphisms.

Theorem 1. *There is a continuous valuation on a Hausdorff topological field whose intrinsic value group is not a power of $\mathbb{Z}$. Consequently Conjecture 00000005754 is false as stated.*

Proof. Take the field and valuation constructed above. Continuity holds for both explicit codomain topologies. Its intrinsic value group is $G \cong \mathbb{Q}$, and Proposition 1 rules out $G \cong \mathbb{Z}^I$ for every I . In particular, for no natural number r does $G \cong \mathbb{Z}^r$ hold. The final universal clause of the conjecture therefore fails for this example, independently of the other clauses. $\square$

6 Formalization and verification scope

The mathematical source is `Counterexample.lean`, using Lean 4.19.0 and Mathlib v4.19.0 pinned to commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The theorem `continuous_valuation_counterexample` combines continuity for both topologies, attainment of every rational value by a nonzero series, the obstruction for every index type, and the explicit negation

$$\neg \exists r \in \mathbb{N}, G \cong (\text{Fin}(r) \rightarrow \mathbb{Z}).$$

The exact definition of G and its relation to w are proved in the preceding named theorems. No unproved rank assignment is assumed.

An independent fresh project build with `lake build`, direct replay of the mathematical source with warnings treated as errors, and a compiled-declaration audit have passed. The audit in `lean/Audit.lean` enumerates every declaration owned by the mathematical module, including compiler-generated declarations: 24 in total. It checks each declaration’s transitive axiom dependencies, type, and direct constants. Only the standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound` occur. `lean/Inspect.lean` displays the exact theorem types, including topology arguments, and checks the field, Hausdorff, and topological division-ring instances. There are no custom axioms, proof placeholders, or untrusted native decision proofs. No auxiliary numerical or symbolic computation is required by the argument.

The supplied `verify.py` rebuilds the authored project in a new directory, replays its sources with warnings treated as errors, compares the complete compiled inventory with the frozen reviewed inventory, and checks the nine dependency revisions and source cleanliness before and after execution. It records actual commands, exit codes, and output. The companion `test_verify.py` contains verifier controls, including compiled examples that must be rejected for an added axiom, an unsafe definition, or an admitted proof. Reproduction commands and the validation records are listed in `README.md` and `VERIFICATION.md`. These are local checks and an independent local semantic review; maintainer acceptance is a separate repository decision.

The supplied original statement was verified against its SHA-256 identity:

`e9168d9270246842f8d39c8e5f8dae0ff58c`
`4723f9957975245e4ab785b720f6`.

The access-provenance and verification logs record the isolated inputs, consulted standard library sources, dependency identities, actual build commands, and their results.